The Interpretation Session: Program ω's First Verdicts

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Companion to the gate note (Revision 1.3), which closes on everything measurable. This note is different in kind, not degree: nothing here is two-implemented in the gate note's sense, because nothing here is a computation. It is the session where the corpus's own theorems were asked what they mean, and where two elegant syntheses each survived with exactly one leg honestly relabeled — which is what this project has learned to call knowing something, rather than having guessed well.

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1. Four questions, posed formally and plainly

The session opened with four interpretation questions, each pairing a formal statement with its plain-language content:

- **Q1**: Does the expanding flow's monotone, off-spine depth-motion constitute *settling*, given it departs the quasi-free family immediately rather than moving along it?
- **Q2**: What does the three-class taxonomy (60/12/12) mean physically — is anchor-class a property of a possible history, or bookkeeping?
- **Q3**: Does the 24-anchor frozen core earn a "memory" reading, and does it give the anchor-supplier question (REG-13) its first dynamical handle?
- **Q4**: Does the live scalar channel settle the division of labor between dynamics and algebra for measurement stability?

2. Q1: resolved, by catching the question's own stale premise

[FINDING] The tension in Q1 was itself an artifact. "Settling as movement through the family" is $(\mathbb{S}, \tau, \omega)$-era doctrine, from when χ *was* the family parameter. Current Foundation demotes the whole equilibrium restriction to `[COMMITMENT, not forced]`; Becoming's own χ_{kin} is defined pointwise, off-spine, needing no family at all — its own text states plainly that "depth was never actually a thermal notion." Once this is seen, the flow leaving the family immediately isn't a strike against settling — it's Becoming's own off-spine prediction, realized.

[FACT] The family retains three legitimate roles, and loses exactly one. It remains the calibration line (depth collapses to one number only at equilibrium, exactly as temperature does); the substrate for the corpus's hardest theorems (disjointness, KMS uniqueness, the type-III verdict are all *about* family members); and the supplier of plausible initial conditions (the flow's own $\nu = 1.5$ start). What it no longer supplies is the rail settling is required to run on. Discriminator 3 is relabeled accordingly: from a settling criterion to a **spine-adherence diagnostic**, measuring how fast a flow leaves the scaffold, not whether it counts as becoming.

[OPEN] One connection worth keeping visible: the anchor that settles cleanly (the classic, generic anchor) carries no frozen core; the anchors with a frozen core (the 24 special ones) wander rather than deepen, in the one contracting-class case tested. No anchor tested so far does both. Whether "settling" ever needs to mean more than depth-motion — whether it needs to mean *carrying an orientation*, which is the reason this whole program exists — remains genuinely unaddressed by Q1's resolution alone.

3. The three-selections filing

[FACT, verified against source] The corpus has exactly three structurally parallel unsupplied selections: the measurement outcome, the orientation/anchor, and the direction/arrow. Each shares the same shape — the framework derives the alternatives, derives what governs them, proves its own incapacity to select, and locates the selection outside the derivable. Verified precisely: the order-robustness argument (Paper III) genuinely holds at every order, not merely as a stated hope; the gate's sign-blindness (this note's companion) is genuine and structural, not a check that happened to pass.

[CORRECTION, first pass] The three are not quite parity in one respect. Measurement carries derived statistics (Gleason–Busch forces the Born-rule form); orientation and direction do not — nothing in the corpus assigns any probability, even a symmetric one, between the two mirror classes or the two ends. The first framing of this ("provably zero derived odds") overreached: "odds" isn't a category that applies to a one-time initial condition at all, rather than applying and returning a null value.

[CORRECTION, second pass — F's, deepening the first] Canonical measures do exist, just none is committed as a sampling law. The wall-null $\rho = \mathrm{Id}/16$ and the Haar average $\langle T_A \rangle = 2 \cdot \mathrm{Id}$ over the anchor locus are both genuine, computed measures over these spaces — the corpus's own explicit "Haar" language for the second is verified present. What's missing is any derivation that the world samples from either. Adopting one as a physical sampling law would itself be a further `[COMMITMENT]`, at the same tier as the equilibrium restriction. **The odds question therefore regresses rather than dissolving or returning zero**: any proposed measure over a selection is itself one more unsupplied selection, one level up.

[INTERPRETATION, flagged for pressure-testing rather than endorsed outright] The three-way parity with established open problems in physics (chirality as unexplained input; the measurement-selection gap; now the cosmological measure problem, in its textbook shape) is a striking pattern.

Given how many syntheses this session has needed to correct on inspection, this is recorded as worth stating once, precisely, and testing hard — not yet adopted as a settled frame.

4. The line-crossing argument: assembled, then corrected twice at the same joint

[ARGUMENT, as first assembled] The disjunction is derivable even where the disjunct is not. Three pieces: the label space is binary (derived); every realized world has a sharp label (claimed via factoriality, $M_\omega = \mathbb{C}1$); nothing derivable supplies which value (the protection theorems). Read together: the framework would prove a selection occurred, prove it binary, and prove it cannot say which way — “the unknowable enters carrying a proof of its own arrival.”

[ERROR, caught and corrected] The middle leg cited the wrong axis. Factoriality ($M_{\omega_\beta} = \mathbb{C}1$) is a proven fact about the β-axis — trivial center within one thermal state’s own algebra. It says nothing about orientation or anchor. What the argument actually needed — an analogous disjointness result across different orientations or anchors — does not exist in the corpus; searched for directly and found nothing.

[FACT, F’s resolution, verified] The gap is not merely open — it is closed, false, in the finite construction, with the true axis named. The wall sector is explicitly finite-mode (confirmed in Phase 5’s own text). On any finite-mode CCR/CAR system, the Shale–Stinespring/Powers–Størmer criterion (quasi-equivalence iff the symbol-difference is Hilbert–Schmidt) is satisfied automatically, since every operator on a finite-dimensional space is trivially Hilbert–Schmidt. **The four orientation vacua are therefore genuinely quasi-equivalent as built — superpositions between them are legal states, not merely unforbidden.** This was true the moment the construction was fixed as finite-mode, before anyone checked it.

[REGISTRY ENTRY, filed] REG-15 — the sharp-realization question. Does the framework’s own construction force a physically realized world to carry a definite orientation/anchor value? **Resolved NO, as a theorem, in any finite truncation** (Shale–Stinespring triviality, exact). **Open in the thermodynamic limit**, where the deciding criterion is named: orientation-sectors become disjoint exactly when the mirror-flip acts with positive density across infinitely many modes. The corpus’s existing infinite-volume constructions live on the bosonic scale line; the fermionic/wall module has not been taken to its own limit — this is the actual, located open work.

[INTERPRETATION, F’s reframing, held at its earned weight] If REG-15 resolves YES in the limit, folium-tier sharpness joins type III, event-coarseness, and thermal disjointness as an infinite-volume phenomenon — the world’s definiteness about which mirror it is would share its source with its indefiniteness about individual events. A genuinely elegant possible unification, explicitly not yet a theorem.

5. The ledger, this session

Species named, both recurring from prior arcs and new: - *Theorem-scope inflation*, second occurrence (the λ^3-split's fixed-W scope; now factoriality's β-axis scope). Countermeasure strengthened per the recurrence: any load-bearing theorem citation in a synthesis quotes the theorem's own statement inline, so an axis mismatch is visible in the same line it's used. - *The elegance tax* (new): the most beautiful synthesis in an exchange reliably carries exactly one seam needing exactly this kind of check — named as an observed regularity, not a criticism of the thinking that produces it. - One precision correction on the citation register itself (the wall-null/four-vacua connection stated as a well-supported inference rather than a verbatim quote, before it entered this note as settled).

6. The third flow: pre-registration confirmed, the taxonomy completes

[FACT, one-implementation, awaiting F's confirmation] The pre-registered prediction landed exactly. The $(0, 16)$ pure-antilinear anchor ($e_1 + e_{12}$, $\operatorname{tr} A = +16$) was integrated from the same quasi-free initial state. Result, checked branch by branch across the full trajectory, not sampled: all 8 distinct symplectic eigenvalues are monotone nondecreasing throughout, no exception. The 4-dimensional frozen core stays exactly at $\nu = 1.5$ (deviation 0.0); every active branch strictly increases. The falsifier — any decreasing active branch — did not fire.

This is deepening with memory, the third character, completing the taxonomy:

Class	Anchor	Frozen core	Behavior
Middle $(4, 12)$	$e_1{+}e_{10}$	No	Deepens, no core
Extreme-low $(12, 4)$	$e_4{+}e_9$	Yes	Wanders
Extreme-high $(0, 16)$	$e_1{+}e_{12}$	Yes	**Deepens with memory**

Two items remain, side by side, neither privileged: independent confirmation of §6's result, and REG-15's thermodynamic-limit half. Q2 and Q4 remain formally open, though Q4's own division of labor ("dynamics writes, algebra remembers") reads as well-supported by everything closed so far, pending the session's explicit adoption.

Created in collaboration with Claude (Anthropic). Interpretation-session record, distinct in kind from the gate note's computational register; all conclusions and any errors remain the author's responsibility.